

A decorative graphic on the left side of the slide, consisting of a network of white lines and small circles on a dark brown background, resembling a circuit board or a stylized tree structure.

Software Engineering

Presented By

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Introduction To Course

- **Course Code:** CSE 0613 - 4121
- **Course Title:** Software Engineering
- **Contact Hours / Week:** 3 Hours (2 classes per week)
- **Credit:** 3.00
- Sessional course included.

The background is a dark brown gradient. In the four corners, there are white line-art illustrations of circuit traces. These traces consist of straight lines of varying lengths and angles, some ending in small open circles, resembling a printed circuit board layout.

Lecture 01

Product

Software (Product)

Software is

- **Instructions** (computer programs) that when executed provide desired function and performance,
- **Data structures** that enable the programs to adequately manipulate information and
- **Documents** that describe the operation and use of programs.

Characteristics of Software

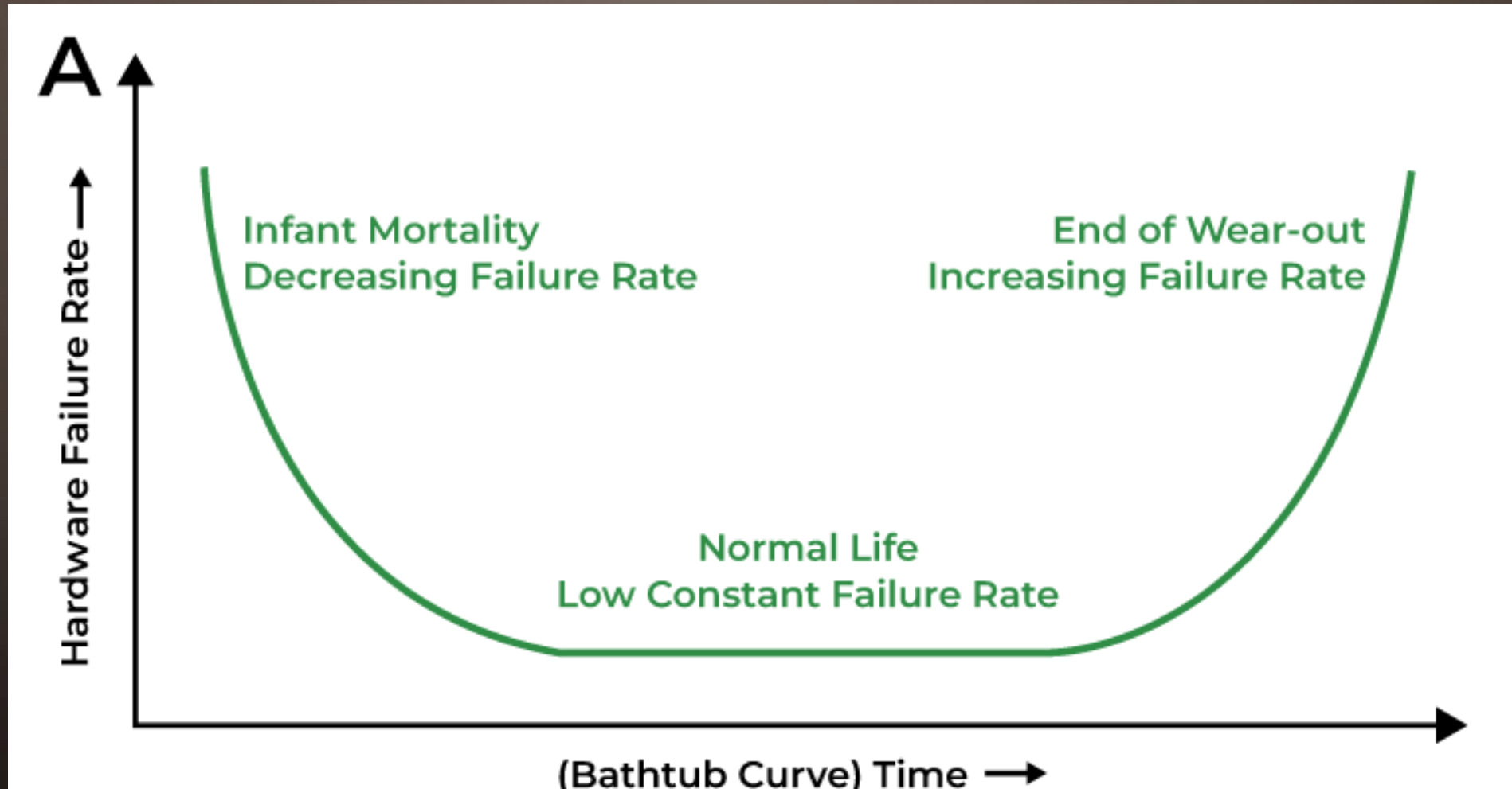
- Software is developed or engineered, it is not manufactured in the classical sense.
 - Two differences exists between software and hardware manufacture.
 - The manufacturing phase for hardware can introduce quality problems that are easily corrected for software.
 - Both activities require the construction of a product , but the approaches are different.

Characteristics of Software

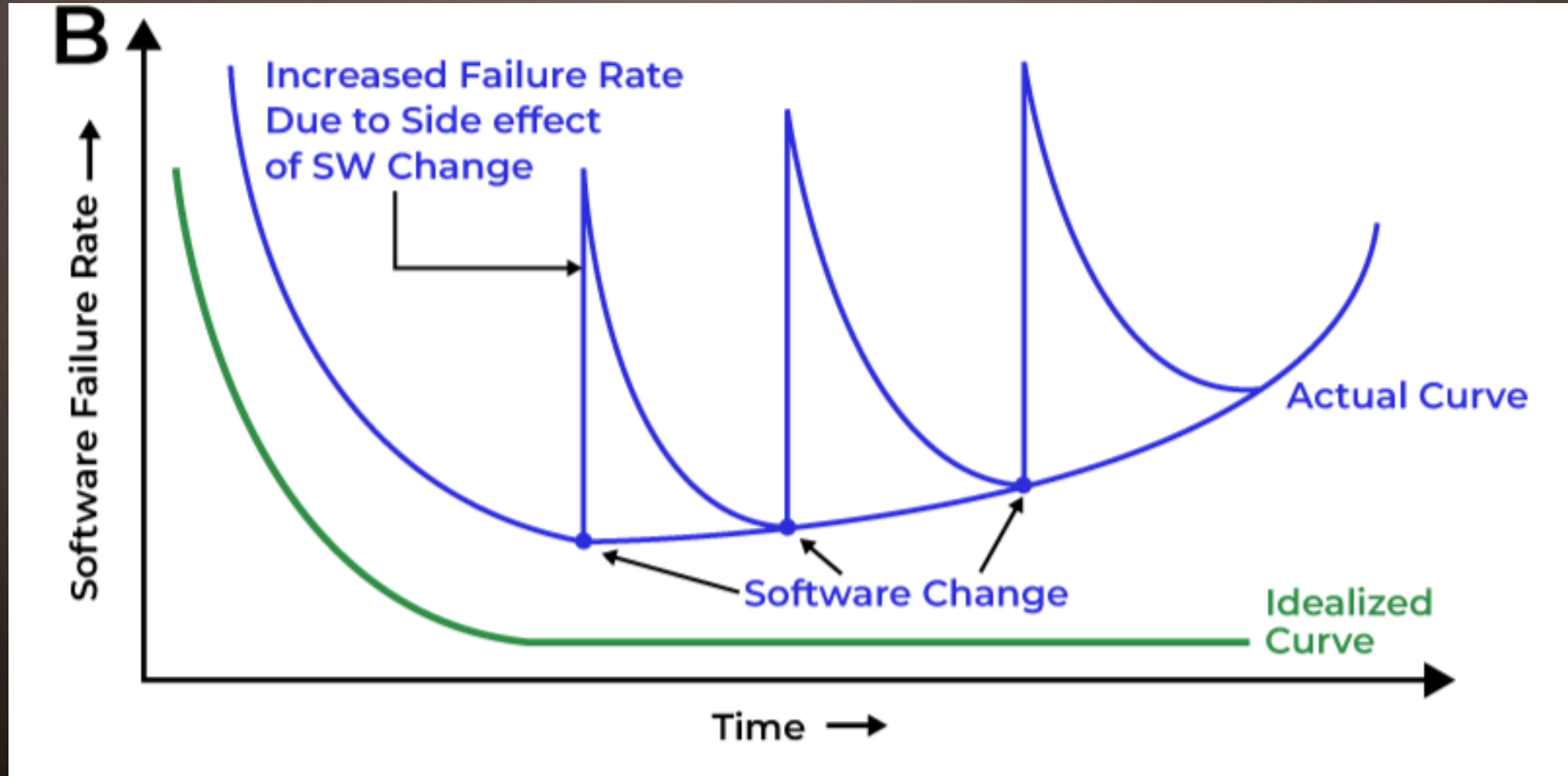
Software doesn't wear out but it does deteriorate.

- Hardware failures are high early in its life, failure rate drops to a steady-state level for some period of time and the failure rate rises again as hardware components suffer from the cumulative affects of dust, vibration, etc.
- For software, undiscovered defects caused high failure rates early in the life of a program, those were corrected without introducing other errors.

Characteristics of Software



Characteristics of Software



Applications of Software

- **System Software**

System software is a collection of programs written to service other programs. System software has heavy interaction with computer hardware, heavy usage by multiple users, etc.

- **Real-time Software**

Software that monitors/analyzes/controls real-world events as they occur is called real time. Elements of real-time software include a data gathering component.

Applications of Software

- **Business software**

Business information processing is the largest single software application area.

E.g. payroll, accounts receivable, inventory, etc.

- **Engineering and scientific software**

Engineering and scientific software have been characterized by "number crunching" algorithms.

E.g. Computer-aided design, system simulation, and other interactive applications.

Applications of Software

- **Embedded software**

Intelligent products have become commonplace in nearly every consumer and industrial market. Embedded software resides in read-only memory and is used to control products and systems for the consumer and industrial markets.

Embedded software can perform very limited and esoteric functions (e.g., keypad control for a microwave oven) Provide significant function and control capability (e.g., digital functions in an automobile such as fuel control, dashboard displays, and braking systems)

Applications of Software

- **Personal computer software**

Word processing, spreadsheets, computer graphics, multimedia, etc, are only a few of hundreds of applications.

- **Web-based software**

The Web pages retrieved by a browser are software that incorporates executable instructions (e.g., CGI, HTML, Perl, or Java), and data (e.g., hypertext and a variety of visual and audio formats).

Applications of Software

- **Artificial Intelligence software**

Artificial intelligence software makes use of non numerical algorithms to solve complex problems that are not amenable to computation or straightforward analysis.

E.g. AI software are pattern recognition, theorem proving and game playing.

Thank You

